

# Theme 1.1 — Nature of Economics and How Markets Work

Edexcel A • A-Level Economics

## 1. The Nature of Economics

**Economics** is the social science that studies how individuals, firms, and governments allocate scarce resources among unlimited wants. It addresses three fundamental questions: *What* to produce? *How* to produce? *For whom* to produce?

Economics is divided into two broad branches:

- **Microeconomics** — the study of individual markets, consumers, and firms (e.g. the price of coffee, a firm's output decision).
- **Macroeconomics** — the study of the economy as a whole (e.g. inflation, unemployment, GDP growth).

### 1.1 Positive and Normative Statements

**Positive statement:** An objective, fact-based claim that can be tested or verified with evidence.  
Example: "UK CPI inflation peaked at 11.1% in October 2022."

**Normative statement:** A subjective, value-based opinion that cannot be proven true or false.  
Example: "The government *should* do more to reduce inequality."

The distinction matters because economic policy debates often mix the two. Recognising which statements are positive (testable) and which are normative (value judgements) helps you evaluate arguments critically.

**Exam tip:** Look for words like “should”, “ought to”, or “it is unfair that” — these signal normative statements. Positive statements can be tested with data, even if the data is not yet available.

## 2. Scarcity, Choice, and Opportunity Cost

### 2.1 The Basic Economic Problem

**Scarcity** exists because resources are finite while human wants are infinite. This creates the basic economic problem: societies cannot produce enough to satisfy everyone’s wants, so choices must be made about how to allocate resources.

The economic problem applies at every level:

- **Individuals** have limited income and must choose between competing wants (e.g. saving vs. spending).
- **Firms** have limited budgets, labour, and capital and must decide what goods or services to produce.
- **Governments** have limited tax revenue and must prioritise spending across health, education, defence, etc.

### 2.2 Opportunity Cost

**Opportunity cost** is the value of the next best alternative foregone when a choice is made. It is the true economic cost of any decision.

Opportunity cost is a central concept that runs throughout the entire A-Level course. Every economic decision involves an opportunity cost because choosing one option means sacrificing another.

#### Examples:

- A student choosing to attend university — the opportunity cost includes the wages they could have earned working full-time for three years, plus the direct cost of tuition.
- A government spending £50 billion on HS2 — the opportunity cost is the next best use of that money, such as building hospitals or schools.

- A firm investing in new machinery — the opportunity cost is the return from putting that money in a bank or investing in marketing.

**Exam tip:** Always identify the *next best* alternative, not just any alternative. Opportunity cost is singular — it is only the one thing you would have done instead.

### 3. Factors of Production

There are four factors of production — the inputs needed to produce goods and services. Each factor earns a return (income) for its owner:

| Factor            | Definition                                                                                 | Return (Income)   | Examples                                                 |
|-------------------|--------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------|
| <b>Land</b>       | All natural resources used in production                                                   | Rent              | Farmland, oil, timber, water, minerals, wind energy      |
| <b>Labour</b>     | The human effort (physical and mental) used in production                                  | Wages / Salaries  | Factory workers, teachers, software developers, surgeons |
| <b>Capital</b>    | Man-made aids to production (not money itself, but physical and human capital)             | Interest / Profit | Machinery, factories, tools, computers, vehicles         |
| <b>Enterprise</b> | The factor that combines the other three; the entrepreneur takes risks and makes decisions | Profit            | Business founders, risk-takers who organise production   |

#### Physical capital vs. human capital

- **Physical capital:** tangible assets like machinery, buildings, and equipment.
- **Human capital:** the skills, knowledge, experience, and education embodied in the workforce. Investment in human capital (e.g. training, education) raises labour productivity.

#### 3.1 Free Goods vs. Economic Goods

**Economic goods** are scarce and have an opportunity cost. Resources must be used to produce them (e.g. a car, a loaf of bread, a haircut).

**Free goods** have no opportunity cost and are not scarce. They exist in unlimited supply (e.g. air, sunlight). In practice, very few goods are truly free — clean air, for instance, requires resources to maintain.

## 4. Production Possibility Frontier (PPF)

**The Production Possibility Frontier (PPF)** shows the maximum possible output combinations of two goods that an economy can produce when all resources are fully and efficiently employed, given the current state of technology.

### 4.1 Key Features of the PPF

- **Points on the curve** — productively efficient; all resources are fully employed.
- **Points inside the curve** — productively inefficient; some resources are unemployed or underutilised (e.g. during a recession).
- **Points outside the curve** — currently unattainable with existing resources and technology, but could become attainable through economic growth.
- The **concave (bowed-out) shape** reflects *increasing opportunity costs*: as more of one good is produced, increasingly more of the other good must be given up, because resources are not perfectly adaptable between uses.

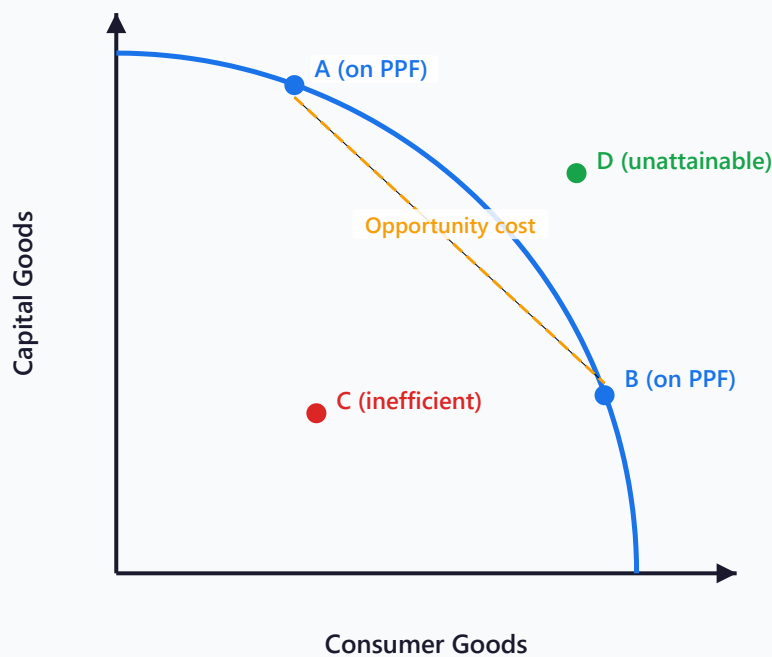


Figure 1: Production Possibility Frontier (quarter-circle shape showing increasing opportunity costs)

## 4.2 Movements Along and Shifts of the PPF

**Movement along the PPF** represents a reallocation of resources between the two goods. Moving from point A to point B means producing more consumer goods but fewer capital goods. The opportunity cost of the extra consumer goods is the capital goods foregone.

**Outward shift of the PPF** represents economic growth — the economy can now produce more of both goods. Causes include:

- Increased quantity of resources (e.g. immigration increasing the labour force, discovery of new natural resources).
- Improved quality of resources (e.g. better education/training raising human capital, technological progress improving capital).

**Inward shift of the PPF** represents a decline in productive capacity. Causes include:

- Natural disasters destroying capital or land.
- Emigration reducing the labour force.
- War or conflict destroying infrastructure.

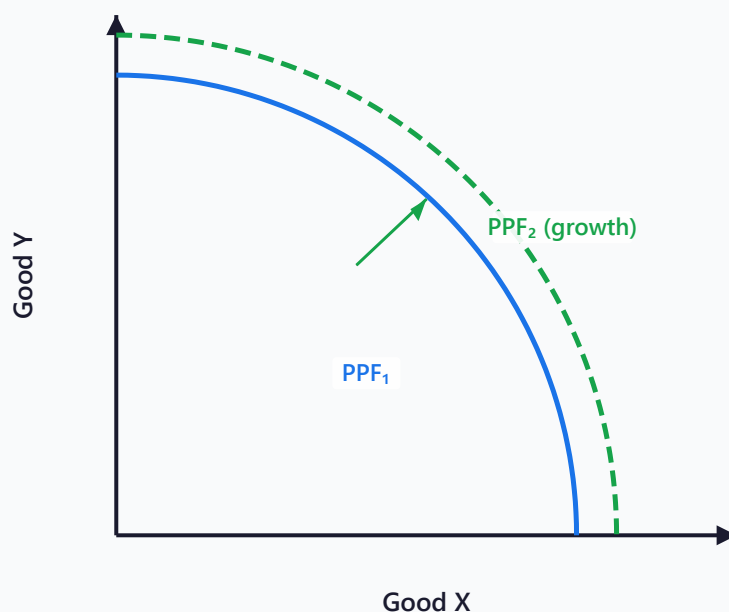


Figure 2: Outward shift of the PPF representing economic growth

### 4.3 PPF and Key Economic Concepts

| Concept                     | Shown on PPF                                                                  |
|-----------------------------|-------------------------------------------------------------------------------|
| Scarcity                    | The boundary of the PPF — cannot produce beyond it with current resources     |
| Choice                      | Choosing a point on the PPF means selecting a particular combination of goods |
| Opportunity cost            | Moving along the PPF — producing more of one good means less of the other     |
| Productive efficiency       | Any point on the PPF curve itself                                             |
| Unemployment / inefficiency | Any point inside the PPF                                                      |
| Economic growth             | An outward shift of the entire PPF                                            |

## 5. Specialisation and Division of Labour

**Specialisation** occurs when individuals, firms, regions, or countries concentrate on producing a limited range of goods or services in which they have a comparative or absolute advantage.

**Division of labour** is a form of specialisation where the production process is broken down into separate tasks, each performed by a different worker.

### 5.1 Advantages of Specialisation and Division of Labour

- **Higher productivity** — workers become skilled at their specific task through repetition, increasing output per worker.
- **Time savings** — workers do not need to switch between tasks or tools, reducing wasted time.
- **Economies of scale** — specialisation allows firms to produce on a larger scale, reducing average costs.
- **Better use of talent** — workers can be allocated to tasks that match their skills and abilities.
- **Facilitates innovation** — workers deeply familiar with a specific process are more likely to find improvements.
- **Enables use of machinery** — breaking production into simple tasks makes it easier to mechanise and automate.

### 5.2 Disadvantages of Specialisation and Division of Labour

- **Worker boredom and demotivation** — repetitive tasks can reduce job satisfaction and increase absenteeism.
- **Structural unemployment** — if demand for a product falls, specialised workers may lack transferable skills.
- **Dependency risk** — if one stage of production fails (e.g. a strike or supply chain disruption), the entire process halts.
- **Loss of flexibility** — workers trained for one narrow task may struggle to adapt to new production methods.
- **Standardisation** — mass production may reduce variety and quality compared to craft production.

### Case Study: Adam Smith and Pin Manufacturing

Adam Smith famously described a pin factory in *The Wealth of Nations* (1776). He observed that one untrained worker alone could make perhaps one pin a day, and certainly not 20. However, when the process was divided into 18 distinct operations (drawing wire, straightening, cutting, pointing, grinding, etc.) among 10 workers, they could produce around 48,000 pins per day — or 4,800 per worker. This dramatic increase in productivity demonstrated the power of division of labour.

## 5.3 Specialisation at Different Levels

- **Individual specialisation:** a surgeon specialises in heart surgery rather than general practice.
- **Firm specialisation:** Apple designs its own chips but outsources their manufacture to TSMC.
- **Regional specialisation:** Silicon Valley specialises in technology; the City of London in financial services.
- **National specialisation:** Saudi Arabia specialises in oil production; Bangladesh in garment manufacturing. This leads to international trade.

## 5.4 Specialisation and the Functions of Money

Specialisation only works if people can exchange what they produce. Barter requires a **double coincidence of wants**, so specialised economies rely on money. Money has four functions:

- **Medium of exchange** — accepted in payment for goods and services, removing the need for barter.
- **Measure of value (unit of account)** — allows the prices of different goods to be compared.
- **Store of value** — can be saved and used to buy goods in the future (weakened by high inflation).
- **Method of deferred payment** — allows goods to be bought now and paid for later (e.g. credit, loans).

# 6. Free Market, Command, and Mixed Economies

## 6.1 Free Market Economy

**Free market economy:** An economic system where resource allocation is determined entirely by the forces of supply and demand, without government intervention. The price mechanism acts as an

"invisible hand" (Adam Smith) to coordinate decisions.

**Key features:**

- Private ownership of factors of production.
- Prices determined by market forces (supply and demand).
- Profit motive drives firms; self-interest drives consumers.
- Consumer sovereignty — consumers determine what is produced through their spending decisions.
- No (or minimal) government intervention.

**Advantages:** efficient allocation via the price mechanism, consumer choice, incentives for innovation and enterprise, no bureaucratic waste.

**Disadvantages:** market failure (externalities, public goods under-provided, inequality), merit goods under-consumed, demerit goods over-consumed, monopoly power.

## 6.2 Command (Planned) Economy

**Command economy:** An economic system where the government or central planning authority makes all major decisions about what, how, and for whom to produce. Resources are state-owned.

**Key features:**

- State ownership of factors of production.
- Central planning body sets output targets, prices, and wages.
- Government decides resource allocation.
- Little or no consumer choice in what is produced.

**Advantages:** can provide public and merit goods, can reduce inequality, can direct resources to priority areas, avoids wasteful duplication.

**Disadvantages:** information problems (planners cannot know all consumer preferences), lack of incentives, bureaucratic inefficiency, shortages and surpluses common, restricted individual freedom.

## 6.3 Mixed Economy

**Mixed economy:** An economic system that combines elements of both free market and command economies. Markets allocate most resources, but the government intervenes to correct market

failures and provide public services.

Most real-world economies are mixed. The UK, for example, has a predominantly market-based system but the government provides the NHS, state education, and welfare; regulates markets; and uses taxation and spending to redistribute income. The degree of government intervention varies across countries and over time.

**Evaluation:** There is no “correct” balance between market and government. The optimal degree of intervention is a normative question that depends on value judgements about equity, efficiency, freedom, and the role of the state.

## 6.4 Key Thinkers: Smith, Marx and Hayek

- **Adam Smith** (*The Wealth of Nations*, 1776) — individuals pursuing their own self-interest are guided by the “invisible hand” of the market to promote the interests of society; he stressed specialisation and the division of labour, and a limited role for the state (e.g. defence, justice, public works).
- **Karl Marx** (*Das Kapital*, 1867) — argued that capitalism exploits workers, because capitalists keep the surplus value created by labour; he predicted that capitalism would be replaced by collective ownership of the means of production, the basis for command economies.
- **Friedrich Hayek** (*The Road to Serfdom*, 1944) — argued that prices convey dispersed information that no central planner could ever collect, so free markets allocate resources better than planning, and that extensive state control threatens individual freedom.

## 7. Demand

**Demand** is the quantity of a good or service that consumers are willing and able to buy at a given price, in a given time period. Both willingness *and* ability to pay must be present — this is known as **effective demand**.

### 7.1 The Law of Demand

The law of demand states that, *ceteris paribus* (all other things being equal), there is an **inverse relationship** between the price of a good and the quantity demanded. As price rises, quantity demanded falls; as price falls, quantity demanded rises.

### Reasons for the downward-sloping demand curve:

- **Income effect:** When the price of a good falls, consumers' real income (purchasing power) rises, enabling them to buy more of the good.
- **Substitution effect:** When the price of a good falls relative to alternatives, consumers switch towards the now-cheaper good and away from substitutes.
- **Diminishing marginal utility:** Each additional unit consumed provides less extra satisfaction, so consumers will only buy more if the price is lower.

## 7.2 Individual and Market Demand

**Individual demand** is the demand of a single consumer. **Market demand** is the horizontal summation of all individual demand curves — at each price, you add together all consumers' quantities demanded.

## 7.3 The Demand Curve



Figure 3: A demand curve — showing movement along the curve when price changes

## 7.4 Movements Along vs. Shifts of the Demand Curve

|  | Movement Along | Shift of the Curve |
|--|----------------|--------------------|
|--|----------------|--------------------|

|                    |                                                                |                                                          |
|--------------------|----------------------------------------------------------------|----------------------------------------------------------|
| <b>Cause</b>       | A change in the good's own price                               | A change in a non-price factor (a "condition of demand") |
| <b>Effect</b>      | Move from one point to another on the <i>same</i> demand curve | The <i>entire</i> demand curve shifts left or right      |
| <b>Terminology</b> | Change in <i>quantity demanded</i>                             | Change in <i>demand</i>                                  |

## 7.5 Determinants of Demand (Shift Factors)

The following non-price factors cause the demand curve to shift:

### 1. Income:

- *Normal goods* — demand increases as income rises (e.g. organic food, holidays). Positive relationship.
- *Inferior goods* — demand decreases as income rises (e.g. value-brand products, bus travel). Negative relationship.

### 2. Price of related goods:

- *Substitutes* (goods in competitive demand) — if the price of a substitute rises, demand for this good increases (e.g. Pepsi price rises → demand for Coca-Cola increases).
- *Complements* (goods in joint demand) — if the price of a complement rises, demand for this good decreases (e.g. petrol price rises → demand for large SUVs falls).

3. **Tastes and preferences:** changes in fashion, advertising, health concerns, social trends (e.g. growing health awareness increases demand for plant-based milk).

4. **Population:** a larger or changing population structure alters total market demand (e.g. an ageing population increases demand for healthcare).

5. **Expectations of future price changes:** if consumers expect prices to rise in the future, current demand increases (e.g. buying before a tax increase).

6. **Interest rates:** lower interest rates reduce the cost of borrowing, increasing demand for goods typically bought on credit (e.g. cars, houses).

7. **Government policy:** changes in taxation, subsidies, or regulation can shift demand.

### Case Study: The Oat Milk Demand Shift

UK sales of oat milk grew rapidly in the late 2010s and early 2020s. This dramatic rightward shift in

demand was driven by several non-price factors: changing consumer tastes (rising veganism and environmental awareness), concerns about dairy farming's carbon footprint, social media influence, and increasing availability in coffee shops. Brands like Oatly capitalised on this trend. The shift in demand means that at every price, more oat milk was demanded; *ceteris paribus*, this raises both equilibrium price and quantity.



Figure 4: A rightward shift of the demand curve (increase in demand)

## 7.6 Rational Decision Making and Alternative Views of Consumer Behaviour

Traditional theory assumes consumers are **rational**: they aim to **maximise utility** (satisfaction), and firms aim to maximise profit. The **law of diminishing marginal utility** states that as more units of a good are consumed, the extra satisfaction from each additional unit falls.

Behavioural economics suggests consumers often do **not** behave rationally, because of:

- **The influence of other people's behaviour** — people copy others (herd behaviour and social norms), e.g. buying a product because friends or influencers have it.
- **Habitual behaviour** — people repeat past choices without comparing alternatives, e.g. staying with the same energy supplier or bank.
- **Consumer weakness at computation** — people find it hard to weigh up complex information such as interest rates, probabilities or long-term costs, so they use rules of thumb and may misjudge value.

## 8. Supply

**Supply** is the quantity of a good or service that producers are willing and able to sell at a given price, in a given time period.

### 8.1 The Law of Supply

The law of supply states that, *ceteris paribus*, there is a **positive (direct) relationship** between the price of a good and the quantity supplied. As price rises, quantity supplied rises; as price falls, quantity supplied falls.

#### Why does the supply curve slope upwards?

- **Profit motive:** higher prices mean higher potential profits, incentivising firms to produce and sell more.
- **New entrants:** higher prices attract new firms into the market.
- **Increasing marginal costs:** as output rises, the cost of producing each extra unit tends to increase (due to diminishing returns in the short run), so firms need a higher price to justify the extra production.

### 8.2 The Supply Curve



Figure 5: An upward-sloping supply curve

### 8.3 Movements Along vs. Shifts of the Supply Curve

|                    | Movement Along                                                 | Shift of the Curve                                       |
|--------------------|----------------------------------------------------------------|----------------------------------------------------------|
| <b>Cause</b>       | A change in the good's own price                               | A change in a non-price factor (a "condition of supply") |
| <b>Effect</b>      | Move from one point to another on the <i>same</i> supply curve | The <i>entire</i> supply curve shifts left or right      |
| <b>Terminology</b> | Change in <i>quantity supplied</i>                             | Change in <i>supply</i>                                  |

### 8.4 Determinants of Supply (Shift Factors)

1. **Costs of production:** if costs rise (e.g. wages, raw materials, energy), supply decreases (shifts left); if costs fall, supply increases (shifts right).
2. **Technology:** improvements in technology lower costs and increase productivity, shifting supply right.
3. **Number of firms in the market:** more firms entering the market increases total supply.
4. **Indirect taxes:** taxes on producers (e.g. VAT, excise duty) raise costs, reducing supply (shift left).
5. **Subsidies:** government payments to producers reduce costs, increasing supply (shift right).
6. **Weather / natural factors:** particularly important for agricultural products (e.g. drought reduces supply of wheat).
7. **Productivity of labour:** higher productivity means more output per worker, shifting supply right.
8. **Related goods in supply:** goods in joint supply (e.g. beef and leather) and competitive supply affect supply decisions.

#### Case Study: UK Housing Market Supply

The UK has faced a persistent housing supply shortage. Despite strong demand, the supply curve for housing has been relatively inelastic (slow to shift right) due to several factors: strict planning regulations and green belt restrictions limit available land; the construction industry faces skills shortages; building materials costs have risen significantly post-Brexit and post-COVID; and large developers may strategically drip-feed housing to maintain prices. The long-standing target of

300,000 new homes a year in England has never been met, with net additions rarely exceeding 250,000; the current government's target is 1.5 million homes over the 2024–29 Parliament. This supply constraint is a key driver of rising UK house prices, illustrating how supply-side barriers can have significant welfare effects.

## 9. Market Equilibrium

**Market equilibrium** occurs where the quantity demanded equals the quantity supplied at a given price. At the equilibrium price ( $P^*$ ), there is no tendency for the price to change — the market "clears".

### 9.1 How Equilibrium is Reached

If the price is **above equilibrium**, there is a **surplus** (excess supply) — quantity supplied exceeds quantity demanded. Producers will reduce prices to sell their unsold stock, and the price falls towards equilibrium.

If the price is **below equilibrium**, there is a **shortage** (excess demand) — quantity demanded exceeds quantity supplied. Consumers compete for the limited supply, bidding the price up towards equilibrium.

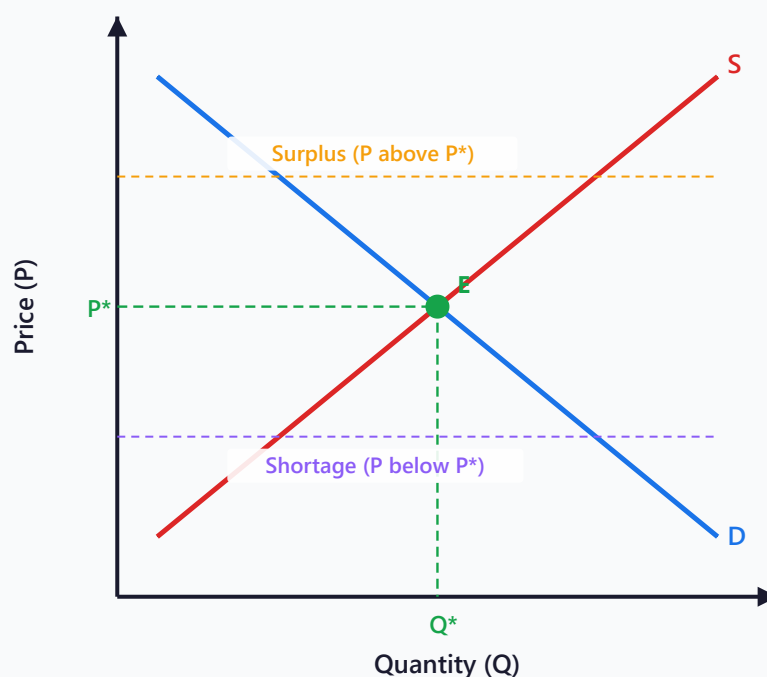


Figure 6: Market equilibrium where supply meets demand

## 9.2 Shifts in Demand and Supply and Effects on Equilibrium

| Change                              | Effect on Price | Effect on Quantity |
|-------------------------------------|-----------------|--------------------|
| Increase in demand (D shifts right) | Price rises     | Quantity rises     |
| Decrease in demand (D shifts left)  | Price falls     | Quantity falls     |
| Increase in supply (S shifts right) | Price falls     | Quantity rises     |
| Decrease in supply (S shifts left)  | Price rises     | Quantity falls     |

When both supply and demand shift simultaneously, the effect on one variable (price or quantity) will be determinate, but the other will be indeterminate (depends on the relative size of the shifts).

## 9.3 Functions of the Price Mechanism

- **Rationing** — when a good is scarce, a higher price reduces quantity demanded so the good goes to those most willing and able to pay.
- **Incentive** — a higher price encourages producers to supply more because it offers higher profits.
- **Signalling** — price changes send information to producers and consumers about where resources are wanted, e.g. rising prices signal firms to enter a market.

Price determination can be applied to **local, national and global markets**, for example local housing, the national labour market and global markets for commodities such as oil.

# 10. Consumer and Producer Surplus

**Consumer surplus** is the difference between what consumers are willing to pay for a good and what they actually pay. It represents the extra utility or satisfaction consumers receive. Graphically, it is the area below the demand curve and above the equilibrium price.

**Producer surplus** is the difference between the price producers receive and the minimum price

they would have been willing to accept. It represents the extra reward to producers above their costs. Graphically, it is the area above the supply curve and below the equilibrium price.

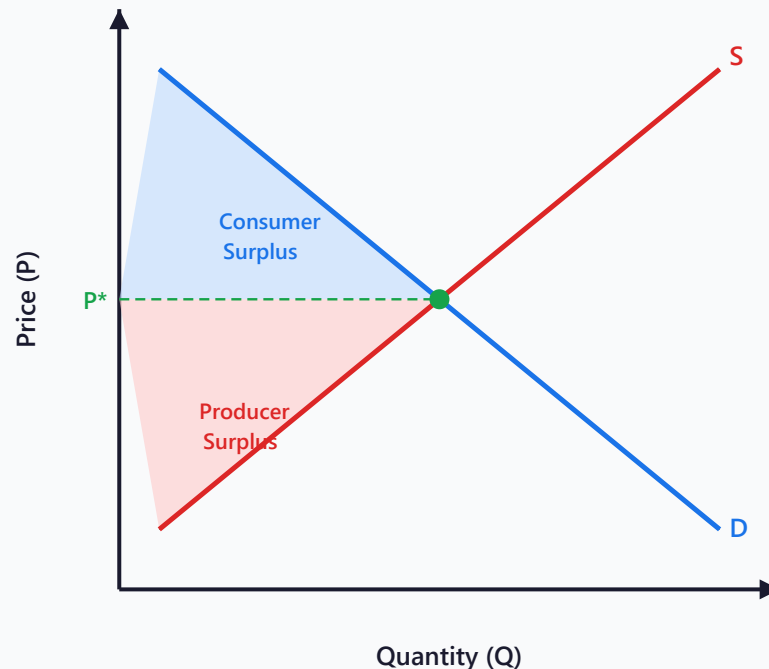


Figure 7: Consumer surplus (blue area) and producer surplus (red area) at market equilibrium

**Total economic welfare** = Consumer surplus + Producer surplus. At the free market equilibrium (where  $D=S$ ), total welfare is maximised and allocative efficiency is achieved. Any output other than  $Q^*$  results in a deadweight loss (welfare loss).

**Exam tip:** When asked how a price change or government intervention affects welfare, draw the new diagram and identify how consumer surplus and producer surplus change. Also consider whether a **deadweight loss** is created.

## 11. Price Elasticity of Demand (PED)

**Price Elasticity of Demand (PED)** measures the responsiveness of quantity demanded to a change in the good's own price.

$$\text{PED} = \% \text{ change in quantity demanded} \div \% \text{ change in price}$$

PED is almost always **negative** because of the inverse relationship between price and quantity demanded (law of demand). We often refer to the absolute value for comparison.

## 11.1 Interpreting PED Values

| Value                  | Term                | Meaning                                                     | Example                                        |
|------------------------|---------------------|-------------------------------------------------------------|------------------------------------------------|
| $ \text{PED}  > 1$     | Price elastic       | Quantity demanded changes proportionately more than price   | Luxury holidays, branded clothing              |
| $ \text{PED}  < 1$     | Price inelastic     | Quantity demanded changes proportionately less than price   | Petrol, salt, insulin                          |
| $ \text{PED}  = 1$     | Unit elastic        | Quantity demanded changes in exact proportion to price      | Theoretical benchmark                          |
| $\text{PED} = 0$       | Perfectly inelastic | Quantity demanded does not change at all when price changes | Life-saving medication (extreme)               |
| $\text{PED} = -\infty$ | Perfectly elastic   | Any price increase causes demand to fall to zero            | Perfectly competitive market (individual firm) |

## 11.2 Determinants of PED

1. **Availability of substitutes:** The more substitutes available, the more elastic demand is, because consumers can easily switch. This is the most important determinant.
2. **Necessity vs. luxury:** Necessities tend to have inelastic demand (people buy them regardless of price); luxuries tend to have elastic demand.
3. **Proportion of income spent:** Goods that take up a large share of income tend to have more elastic demand (consumers are more sensitive to price changes).
4. **Time period:** Demand tends to be more elastic in the long run because consumers have more time to find alternatives and adjust their behaviour.
5. **Habit / addiction:** Addictive goods (cigarettes, alcohol) tend to have inelastic demand.
6. **Brand loyalty:** Strong brand loyalty makes demand more inelastic.

7. **Definition of the market:** Narrowly defined markets (e.g. Coca-Cola) have more elastic demand than broadly defined markets (e.g. soft drinks).

### Case Study: Petrol Price Inelasticity

Petrol is a classic example of a price-inelastic good. When global oil prices spiked during 2022 (due to the Russia-Ukraine conflict), UK petrol prices exceeded £1.90 per litre — a record high. Yet petrol consumption fell only modestly. This is because petrol has few short-run substitutes for most drivers (particularly those in rural areas or commuting), it is a necessity for daily life, and switching to electric vehicles requires a large upfront investment. However, in the long run, PED becomes more elastic: the 2022 price shock accelerated the shift towards EVs, with UK battery-electric car registrations rising by around 40% that year. This illustrates how the time period is a crucial determinant of PED.

## 11.3 PED and Total Revenue

The relationship between PED and total revenue ( $TR = P \times Q$ ) is crucial for firms' pricing decisions:

| PED                                 | Price Increase                                        | Price Decrease                                       |
|-------------------------------------|-------------------------------------------------------|------------------------------------------------------|
| <b>Elastic</b> ( $ PED  > 1$ )      | TR <b>falls</b> (large drop in Q outweighs higher P)  | TR <b>rises</b> (large rise in Q outweighs lower P)  |
| <b>Inelastic</b> ( $ PED  < 1$ )    | TR <b>rises</b> (small drop in Q, higher P dominates) | TR <b>falls</b> (small rise in Q, lower P dominates) |
| <b>Unit elastic</b> ( $ PED  = 1$ ) | TR <b>unchanged</b>                                   | TR <b>unchanged</b>                                  |

**Exam tip:** If a firm has price-inelastic demand, it should raise prices to increase revenue. If demand is elastic, it should lower prices (or offer sales) to boost revenue. This is why luxury brands rarely discount, while supermarkets run frequent price promotions on branded products with many substitutes.

## 11.4 PED and Taxation

The incidence (burden) of an indirect tax depends on PED:

- If demand is **inelastic**, the consumer bears most of the tax burden because they are unresponsive to price changes — the producer can pass the tax on.
- If demand is **elastic**, the producer bears most of the tax burden because they cannot raise the price without losing a large proportion of sales.

This is why governments tax price-inelastic goods (petrol, cigarettes, alcohol) heavily — they generate more revenue and demand does not fall much.

## 12. Income Elasticity of Demand (YED)

**Income Elasticity of Demand (YED)** measures the responsiveness of quantity demanded to a change in consumer income.

$$\text{YED} = \% \text{ change in quantity demanded} \div \% \text{ change in income}$$

### 12.1 Interpreting YED Values

| Value                | Type of Good                    | Meaning                                            | Examples                                            |
|----------------------|---------------------------------|----------------------------------------------------|-----------------------------------------------------|
| $\text{YED} > 1$     | Superior / luxury good (normal) | Demand rises proportionately more than income      | Designer clothing, fine dining, sports cars         |
| $0 < \text{YED} < 1$ | Necessity (normal)              | Demand rises, but proportionately less than income | Bread, toothpaste, electricity                      |
| $\text{YED} < 0$     | Inferior good                   | Demand falls as income rises                       | Value-brand foods, second-hand clothing, bus travel |

**Key point:** The *sign* of YED tells you whether the good is normal (positive) or inferior (negative). The *magnitude* tells you the degree of responsiveness.

### 12.2 Significance of YED

- **Business strategy:** Firms selling luxury goods ( $YED > 1$ ) will see demand grow rapidly during economic booms but fall sharply during recessions. They may need to diversify or hold cash reserves. Firms selling necessities ( $0 < YED < 1$ ) have more stable demand across the economic cycle.
- **Government revenue:** During economic growth, tax revenues from luxury goods increase disproportionately.
- **Structural change:** As economies grow, demand shifts from agriculture (low YED) towards services (higher YED). This explains why developed economies are service-dominated.
- **Primary commodity producers:** Developing countries exporting primary goods with low YED may face worsening terms of trade as global incomes grow — demand for their exports grows slowly while demand for manufactured imports grows quickly.

**Evaluation:** YED values are not fixed — they change over time and vary between countries and income groups. A good that is a luxury at low income levels may become a necessity as income rises (e.g. smartphones). YED also depends on the specific product within a category.

## 13. Cross Elasticity of Demand (XED)

**Cross Elasticity of Demand (XED)** measures the responsiveness of quantity demanded of Good A to a change in the price of Good B.

$$XED = \% \text{ change in quantity demanded of Good A} \div \% \text{ change in price of Good B}$$

### 13.1 Interpreting XED Values

| Value                   | Relationship | Meaning                                   | Examples                                            |
|-------------------------|--------------|-------------------------------------------|-----------------------------------------------------|
| $XED > 0$<br>(positive) | Substitutes  | Rise in price of B increases demand for A | Butter & margarine; Uber & taxis; Netflix & Disney+ |
| $XED < 0$<br>(negative) | Complements  | Rise in price of B decreases demand for A | Printers & ink; cars & petrol; phones & cases       |

|                       |                 |                                          |                     |
|-----------------------|-----------------|------------------------------------------|---------------------|
| XED = 0 or close to 0 | Unrelated goods | Price of B has no effect on demand for A | Bread & televisions |
|-----------------------|-----------------|------------------------------------------|---------------------|

The **magnitude** of XED indicates the strength of the relationship. A very high positive XED means the goods are very close substitutes; a very negative XED means they are strong complements.

## 13.2 Significance of XED

- **Pricing strategy:** If a firm knows its product has a high positive XED with a competitor's product, it must be cautious about raising prices (customers will switch easily). Conversely, if XED is low, the firm has more pricing power.
- **Competition policy:** Regulators (like the CMA) use XED to define market boundaries. If two goods have a very high positive XED, they are in the same market. This matters for assessing mergers and market dominance.
- **Complementary pricing:** Firms may sell one complementary good cheaply (printers, game consoles) to boost demand for the profitable complement (ink cartridges, games).

# 14. Price Elasticity of Supply (PES)

**Price Elasticity of Supply (PES)** measures the responsiveness of quantity supplied to a change in price.

$$\text{PES} = \% \text{ change in quantity supplied} \div \% \text{ change in price}$$

PES is always **positive** (because of the positive relationship between price and quantity supplied).

## 14.1 Interpreting PES Values

| Value   | Term                 | Meaning                                                   |
|---------|----------------------|-----------------------------------------------------------|
| PES > 1 | Price elastic supply | Quantity supplied changes proportionately more than price |

|                |                        |                                                                                               |
|----------------|------------------------|-----------------------------------------------------------------------------------------------|
| PES < 1        | Price inelastic supply | Quantity supplied changes proportionately less than price                                     |
| PES = 0        | Perfectly inelastic    | Quantity supplied does not change (vertical supply curve, e.g. land, short-run stadium seats) |
| PES = $\infty$ | Perfectly elastic      | Any amount can be supplied at the same price (horizontal supply curve)                        |

## 14.2 Determinants of PES

1. **Spare capacity:** If firms have unused capacity, they can increase output quickly in response to price rises → PES is more elastic.
2. **Time period:**
  - *Momentary run:* supply is perfectly inelastic (e.g. fish already caught at a market).
  - *Short run:* supply is relatively inelastic (firms can adjust variable factors but not fixed factors).
  - *Long run:* supply is more elastic (firms can adjust all factors, build new factories, enter/exit the market).
3. **Factor mobility:** The easier it is to switch factors of production to produce a good, the more elastic supply is.
4. **Ability to store stock:** If a good can be stored (e.g. tinned food, electronics), producers can release stock when prices rise, making supply more elastic. Perishable goods (fresh flowers, fish) have more inelastic supply.
5. **Availability of raw materials:** If inputs are readily available, supply can increase quickly.
6. **Nature of the product:** Agricultural products often have inelastic supply due to growing seasons (cannot produce more wheat instantly). Manufactured goods tend to have more elastic supply.

## 14.3 PES and Market Outcomes

PES has important implications for how markets respond to demand changes:

- If supply is **inelastic**, an increase in demand leads to a **large price rise** but only a **small quantity increase**.
- If supply is **elastic**, an increase in demand leads to a **small price rise** but a **large quantity increase**.

### Case Study: UK Housing Market and PES

The UK housing market is a prime example of price-inelastic supply. When demand for housing increased sharply (driven by population growth, low interest rates, foreign investment, and Help to Buy), house prices soared — the average UK house price rose from around £170,000 in 2010 to over £280,000 by 2024. Yet supply responded sluggishly: planning restrictions, land scarcity (especially in London and the South East), labour shortages in construction, and long build times mean that PES for housing is very low in the short to medium term. The result is that almost all of the demand increase feeds into higher prices rather than more homes being built. This illustrates why inelastic supply amplifies the price effect of demand shifts.

## 15. Summary Table — All Elasticities

| Elasticity | Formula                                | Sign                               | Key Question                                         |
|------------|----------------------------------------|------------------------------------|------------------------------------------------------|
| <b>PED</b> | $\% \Delta Q_d \div \% \Delta P$       | Negative (usually)                 | How responsive is demand to the good's own price?    |
| <b>YED</b> | $\% \Delta Q_d \div \% \Delta Y$       | + (normal) or – (inferior)         | How does demand respond to income changes?           |
| <b>XED</b> | $\% \Delta Q_d(A) \div \% \Delta P(B)$ | + (substitutes) or – (complements) | How does demand for A respond to price changes in B? |
| <b>PES</b> | $\% \Delta Q_s \div \% \Delta P$       | Positive                           | How responsive is supply to the good's own price?    |

### Evaluation points for any elasticity question:

- Elasticity values are estimates and can change over time — they are not fixed constants.
- The time period matters: most goods become more elastic in the long run.
- Data quality: elasticity estimates depend on accurate price, quantity, and income data.
- Ceteris paribus may not hold in practice — multiple factors change simultaneously.
- Elasticity varies along a linear demand/supply curve (it is not the same as the gradient).

**Exam tip:** In essays and data-response questions, always apply elasticity concepts to the specific context given. Do not just define elasticity — explain what it means for the particular market, firm, or policy being discussed.